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MANUFACTURING PROCESSES

As a mechanical design engineer, I treat design for manufacturing as paramount: a design is only as good as the process that can produce it. I spent roughly 2.5 years as a manufacturing engineer at a premier plant for the world's leading metal-cutting tool company, and I machined parts and build my own prototypes in my former employer's shop. The hands-on habit started early — I grew up in the garage as the son of an engineer, and I owned and operated an LLC handyman business through high school and part of university. Whenever schedules and regulations allow, I personally build, test, and troubleshoot my prototypes end-to-end, including the labor, planning, and documentation (and, admittedly, the frequent troubleshooting).

The list below was compiled from memory and omits the most basic processes (hand drills, hammers, and the like). It covers processes I have personally performed, designed for, or witnessed in production.

Designed for and physically performed	Designed for and witnessed in production	Finishing and post-processing
Machining • CNC machining — programming (CAM) and running: milling, turning, drilling, tapping, grinding • Manual mill and manual lathe • Drilling, reaming, boring • Surface grinder, bench grinder, angle grinder • Most any form of cutting of most materials Fabrication and assembly • Welding (mostly arc and stick) • Press brake, hydraulic and arbor press • Deburring • Wiring and controls (Arduino, Raspberry Pi, PLC, myriad controllers) • Any and every hand tool Additive • 3D printing — FDM, SLA, and SLS; printer maintenance and modification • Materials: PLA, ABS, TPU, TPE, PETG, PA-CF, PC, resins, PA12 and PA6 powders • Silicone molding of prototype parts Inspection • CMM and other dimensional inspection	Metal forming and joining • Sheet metal — bending, stamping, deep drawing, roll-forming • Extrusions • Thread forming • Weldments (fillet weld sizing per AWS D1.1) • Brazing • Piping Casting and molding • Die casting and investment casting • Injection molding and overmolding • Thermoforming Material removal • Wire EDM • Laser cutting Additive (production) • DMLS, DLS, SLS, MJF / PJF Other • Heat treatment • Insulation	Coatings and plating • Powder coating • Painting • Anodizing • Zinc-flake coating (Geomet, ISO 10683) • Hot-dip galvanizing • Epoxy coating Surface preparation • Sand / media blasting • Sanding • Acetone bath (vapor smoothing) Other • Autoclaving